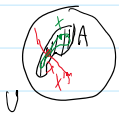
Lecture 9 - 24/06

Sunday, July 5, 2026 8:36 AM

trapping region



a closed shape that at its borders trajectories can only move inwards - never leave

$\varphi(t;U) = \{\varphi(t,x_0) \mid x_0 \in U\} \in \text{interior}(U) \quad t \geq 0$

$\boxed{A = \bigcap_{t\geq 0} \varphi(t;U)}$ - attractor for U

$x \in A$, means $\forall t \geq 0 \quad x \in \varphi(t;U) \; ; \; \exists y \in U : x = \varphi(t,y) \; ; \; \varphi(-t;x) \in U$

so for a trajectory that reaches $x \in A$, no matter how far you go back, stays in U since that is true for every point on that trajectorie, so every point must be inside A, so the whole trajectory is fully inside A.

this is not the whole U, for many points, going back in time leaves U

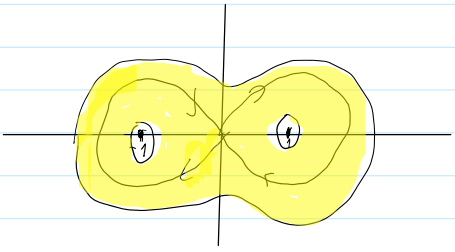
lets look at previous example attractors -

$\dot{x} = y$

$\dot{y} = x - 2x^3 + x(x^2 - x^4 - y)$

$L(x,y) = \frac{-x^2 + x^4}{2} + \frac{y^2}{2}$

$L' = -2y^2 L$



take U, an outer cycle without the FPs

lets call the figure 8 Γ (Γ^+ for $x>0$, Γ^- for $x<0$)

remember, things outside the Γ shrink to it, so U is a trapping region

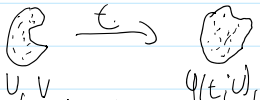
$A = \Gamma$ is invariant set (also Γ^+, Γ^-)

what if U doesn't exclude FP ? $A = \Gamma \cup \{FP_1\} \cup \{FP_2\}$

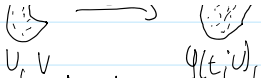
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if we have region U of volume V

define $\varphi(t;U)$ (where U moves after t) and its volume is $V(t)$



for every point $\dot{\vec{x}} = \vec{F}(\vec{x})$

for every point $\vec{x} = \vec{F}(\vec{x})$ 

So the volume depends on the movment of the border
it it goes away, it expands, V inc, if it goes inwards, it shrinks, V dec

if the shape is U, its circumference is ∂U
take an area on the border A, it has a normal and F vector field (what it moves)

$$\dot{V} = \int_{\partial U} (\vec{F} \cdot \vec{n})\, dA = \int_U \underbrace{\nabla \cdot F(\vec{x})}_{\| \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \dots = \sum_i \frac{\partial F_i}{\partial x_i}} dx$$

for 1D, A ——— B it will be

$$\int_A^B \frac{\partial F(x)}{\partial x} dx = F(B) - F(A)$$

it $\sum_i \frac{\partial F_i}{\partial x_i} = C$ constant then $\dot{V} = cV$

chaotic systems - Lorenz

$\dot{x} = -\sigma x + \sigma y$
$\dot{y} = rx - y - xz$
$\dot{z} = -bz + xy$

we assum $\sigma, r, b > 0$

FPs
$\dot{x} = 0, \ y = x$
$\dot{z} = 0, \ z = \frac{x^2}{b}$
$\dot{y} = 0, \ (r-1)x - \frac{x^3}{b} = 0$

fix $\sigma = 10, \ b = \frac{8}{3}$
$r = 28$

$x = 0 \Rightarrow (0,0,0)$
$x \neq 0 \Rightarrow x = \pm\sqrt{b(r-1)}$, if $r > 1$, $\left(\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1\right)$
$z = r-1$

$$DF = \begin{pmatrix} -\sigma & \sigma & 0 \\ r-z & -1 & -x \\ y & x & -b \end{pmatrix}$$

$$DF(0,0,0) = \begin{pmatrix} -\sigma & \sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{pmatrix}$$

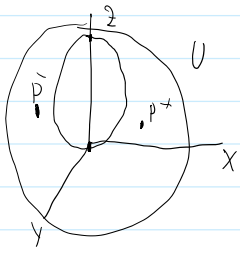
$\lambda_3 = -b < 0$
$\lambda_{s1} = \ < 0$
$\lambda_u = \frac{-(\sigma+1) + \sqrt{(\sigma+1)^2 + 4\sigma(r-1)}}{2}$

for $r < 1$, $(0,0,0)$ is the only FP and $\lambda_u < 0$. it is globally stable
for $r > 1$, $\lambda_u > 0$, which is unstable saddle point

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$$L_2(x,y,z) = \frac{x^2+y^2+(z-r-\sigma)^2}{2}$$

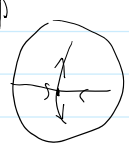
You can prove there is some ellipse where $L_2'(x,y,z) = 0$
and some sphere U around it that is a trapping region, $L_2' \leq 0$



the sphere U has some volume
we can look at $V(t)$

$$\dot{V} = \int_{V(t)} \nabla F \, dx = (-\sigma - 1 - b) \cdot V$$

$$V(t) = V_{sp} \, e^{-(\sigma+1+b)t} \xrightarrow[t\to\infty]{} 0$$

note: a unstable manifold of a saddle FP
that is within a trapping region
is is in the attractor A

why: if we go forward in time, it is trapped
if we go back, it only goes to FP, which is inside U

definition: sensitive dependence on initial conditions

let x_0 be some initial condition
$\exists r > 0, \ \forall \delta > 0, \ \exists y_0 \in B_\delta(x_0), \ \exists \tau > 0$

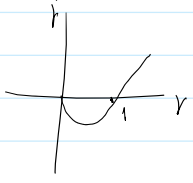
$$\| \varphi(\tau; y_0) - \varphi(\tau; x_0) \| \geq r$$

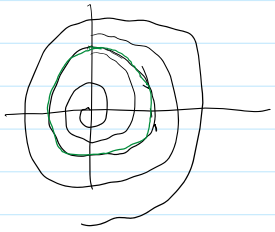
in words, for any small x_0, there is a y_0 that can be as close as δ, after some time τ
they will be as far as r

for lorenz attractor, every point has this property

lets look at trivial examples

$$\dot{r} = r(-1+r^2)$$
$$\dot{\theta} = 1$$





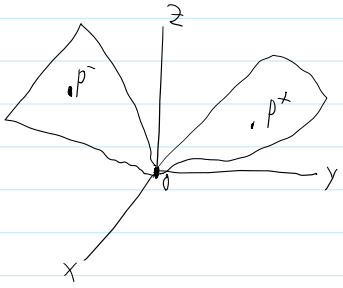
the attractor here is an unstable orbit - the unit cycle
- within it, its not sensitive, points converge to 0
- take x_0 on it, if you can pick y_0 out of it, it is sensitive
- if you can only pick y_0 on the circle, it isn't, so the usual case the atractor is not sensitive

a chaotic system is when we have sensitivity on the attractor

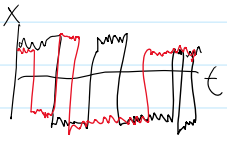
the attractor is not a system

a chaotic system is when we have sensitivity on the attractor

in lorenz the the attractor has 2 sides and flows move between them.
we cannot know how long they stay in each
so take X_0, Y_0 in one, after some time, one will jump to the other side, while the other one doesnt

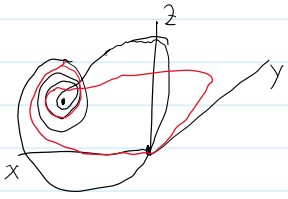


looking at just x over time



another classical example - oscillor attractor

$\dot{x} = -y - z$
$\dot{y} = x + zy$
$\dot{z} = b + z(x -)$

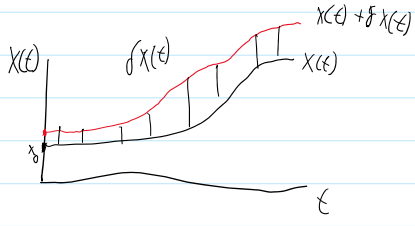


so it spirals inwards, then escapes and comes back to the spiral
we dont know how many cycles around the spiral it will do before escaping

for $\dot{X} = F(X)$ FP X^*
we can linearize if y is close to X^*
$x(t) = X^* + y(t)$
$\dot{y} = DF(X^*) \cdot y$

we can do this for any point, not just FP

$X(t) = \varphi(t, X_0)$
$X(t) + \delta x(t) = \varphi(t, X_0 + \delta x)$, in 1D $\dot{x} = F(x)$



so $\delta x(t)$ is the diviation

$(\dot{X} + \delta\dot{x}) = F(x + \delta x)$
$\dot{x} + \delta\dot{x} = F(x(t)) + \delta x \cdot F'(x(t))$
$\delta\dot{x} = \underbrace{F'(x(t))}_{\alpha(t)} \cdot \delta x(t) = \alpha(t) \cdot \delta x(t)$

this is a linear system

this assumes $\delta x(0)$ is very small

while it is a linear system, which should be simple, you need to solve the full trajectory in order

while it is a linear system, which should be simple, you need to solve the full trajectory in order to get $\alpha(t)$

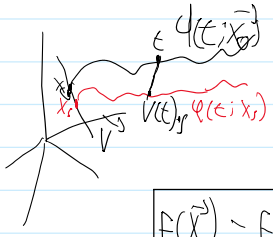
so if you look at $\dot{X} = F(X) \longrightarrow X(t)$

$\delta\dot{X}(t) = \underbrace{F'(X(t))}_{\alpha(t)} \cdot \delta x(t)$

and look at the sign of $\alpha(t)$, if its mostly positive, you have a chaotic system

for the general case - we have $\varphi(t; \vec{X_0})$ and $\vec{V}$ as direction

take $\vec{X_s} = \vec{X_0} + s \cdot \vec{V}$



after time t, we have $V(t)$, the vector between $\varphi(t; X_0)$ and $\varphi(t, X_s)$

$F(\vec{X}) = F(\vec{X_0}) + DF \cdot \delta\vec{X_0}$ near X_0

$\dot{X}(t) = F(X)$

$(X(t) + V(t)\cdot s) = F(X(t) + \dot{V}(t)s) = F(X(t)) + F(\dot{V}(t)\cdot s) = F(X(t)) + s \cdot DF(X(t)) \cdot \vec{V}(t)$

$\dot{V}(t) = \underbrace{DF(X(t))}_{A(t)} \cdot V$

$A(t)$ - a matrix that depends on t, need to find $X(t)$ for it

so this tells us how $v(t)$ vector rotates and change size

we care about $\|V(t)\|$

lets assume $\|V(t)\| \sim e^{\ell t}$

then we can say $\frac{\log\|V(t)\|}{t} \xrightarrow[t\to\infty]{} \ell$ this is a Lyapanove exponent

there cannot be more than n Lyapanov exponents (even if you can take infinite $\vec{V}$)

$\ell_1, \dots, \ell_n$

1 of these is 0, that happens when $\vec{V}$ is the direction of the movement, its basically just like moving ahead in time

if any of $\ell_1, \dots, \ell_n$ is >0, then its a chaotic system, theres a direction which makes the volume expand.

we need to look at $\sum \ell_i$, that controles if the volume shrinks or expands

for lorenz system, $\ell_1 < 0$, $\ell_2 = 0$, $\ell_3 > 0$ while $\sum \ell_i < 0$

picking a random $\vec{V}$ will almost always give the largest ℓ_i (hitting the small one is like hitting the stable manifold of a saddle point, unlikely by accident)

picking a random $\vec{v}$ will almost always give the largest λ_i (hitting the small one is like hitting the stable manifold of a saddle point, unlikely by accident)

the starting x_0 doesn't matter in a chaotic attractor, they'll give the same values

after finding the largest one, you can use it to find the next (gram schmit)